Fiscal Year:	FY 2006	**Task Last Updated:**	FY 10/19/2006
PI Name:	Weichman, Peter B Ph.D.		
Project Title:	Theoretical investigations of equilibrium and nonequilibrium superfluid dynamics with application to future ground-based and microgravity experiments		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Burlington	**State:**	MA
Zip Code:	01803-5024	**Congressional District:**	6
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
Start Date:	04/01/2004	**End Date:**	09/30/2007
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: No cost extension to 9/30/2007, perhaps beyond, per PI (10/06)		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	GSN 4497		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock the mysteries of the universe obscured by Earth's gravity		

Task Description:	Various aspects of the theory of equilibrium and nonequilibrium superfluidity will be considered with special emphasis on properties relevant to proposed and planned future microgravity experiments, and their ground-based precursors. Of special interest are steady equilibrium and nonequilibrium interfacial states, in which gravity (equilibrium) and heat current (nonequilibrium), or a combination of the two, supports coexistence of superfluid and normal phases with a well defined interface between them. Approximate mean field and (where feasible) more accurate renormalization group calculations will be performed (a) to extend present understanding of the steady state structure of these interfaces, and (b) to understand their near-steady state dynamics under the influence of various experimental probes, such as second sound. Also of interest are further investigations into the nature of the so-called self-organized critical (SOC) state, which supports a finite temperature gradient in the superfluid phase by generating a stream of vortex lines propagating across the direction of heat flow. I will investigate the fluctuations and spatial correlations in this state, its associated nonequilibrium specific heat, and the effects of a superfluid-SOC interface on measurements of the heat capacity at constant heat current.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	I have made substantial progress on a number of tasks, with manuscripts in final stages of preparation for publication. (1) In collaboration with Robert Duncan's group and University of New Mexico, we have computed the magnitude of the temperature gradient induced by a heat current in the superfluid phase by Earth's gravity in the absence of dissipation. The gradient is induced by the variation of the superfluid density with height in the cell, and it was originally thought that this mechanism might at least partially explain the size of the experimentally observed gradient. However, the calculated value turns out to be much smaller, and we conclude that the observed value must be dissipation-induced, for example by vortex ring nucleation. A manuscript entitled "Temperature profiles of hydrodynamic origin in counterflowing helium-II near T_lambda" is almost ready to be submitted for publication. (2) In previous work, my collaborators and I considered the large-scale, long-time dynamics of the heat-current-induced superfluid-normal interface. We showed that an interfacial second sound mode exists, in which an initial perturbation to the interface position will propagate at a well defined speed C. This prediction has never been experimentally verified. I have now proposed an "interfacial spectroscopy" experiment which should show an unambiguous signature of the interface mode. Thus, if a second sound pulse is propagated toward, and scattered by the interface, it is predicted to be almost entirely absorbed when the horizontal component of its phase speed matches C. The observed reflected amplitude then shows a strong dip as a function of pulse frequency, with width proportional to interfacial dissipation constant D. A manuscript entitled "Spectroscopy of a nonequilibrium superfluid-normal interface" is almost ready to be submitted for publication to Physics Review Letters. SUPPLEMENTAL RESEARCH PROGRESS FOR 2006 The manuscript entitled "Spectroscopy of a nonequilibrium superfluid-normal interface" has now been published in Physical Review Letters 96, 135301 (2006). Since my 2006 funding increment did not arrive until September 2006, I have been unable to work on this project essentially since that paper was submitted for publication in January 2006, and so I have no new progress to report. Now that the funding is in, I will soon begin pursuing some new ideas on noise and correlations in the self-organized critical state of non-equilibrium superfluid He-4.
Bibliography Type:	Description: (Last Updated: 01/12/2021)
Articles in Peer-reviewed Journals	Weichman PB. "Second-sound spectroscopy of a nonequilibrium superfluid-normal interface." Phys Rev Lett. 2006 Apr 7;96(7):135301. , Apr-2006